

Technical Document #708: Database Systems - Comprehensive Overview

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Database indexing improves query performance by creating data structures that allow faster data retrieval.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Database replication copies data across multiple servers.
- Database monitoring tracks performance metrics and identifies bottlenecks.